

Initial Individual Project Definition 25/26 (7CCSMPRJ)

Describe here your initial thoughts about your project. In this "project proposal", I am looking for a project that is aligned to your topic, sensible and has the potential to deliver something of value, that is sophisticated enough to allow you to earn a good mark for your project, that interests and motivates you and allows you to achieve your objectives.

1. What is a potential TITLE of your project? Actually your title on EMS is a generic TOPIC and you need to refine it to your interests and questions that you want to answer. * 

Enhancing AI Education Engagement Through Puzzle-Based Learning: Integrating IBM

2. What is the problem that you are trying to solve? Before thinking of a solution, find or propose problem(s) or research question(s) that need an answer. You will develop this more throughout the project, but it is essential to think of the problem first before jumping into a solution. * 

The problem is that online learning has low engagement. It is easy to pause or stop, and hard to concentrate on. We need to keep students engaged and motivated to complete the course.

Currently, we have games like "while True: learn()" to make players learn AI knowledge, and we have LLM to teach people knowledge as well. However, there is limited research that merges these two approaches together.

Research Questions:

1. Can a game-based approach effectively teach AI concepts from IBM

3. What is the BACKGROUND/context for this project? (1 paragraph is plenty) * 

AI knowledge is becoming essential today, but the completion rate of online courses is quite low. LLM can teach people knowledge, but the hallucination problem is an issue, and the structure of teaching is a problem as well. Game-based teaching has a significant effect on learning knowledge, but the knowledge is limited to what developers set. If we merge these two approaches, we can use the game setting to limit the hallucination and use LLM to extend knowledge that is limited in the game setting. These two approaches

4. What solution and SOFTWARE do you hope to deliver by the end of the project?



The software I hope to deliver by the end of the project is an educational game made with Unity, similar to "while True: learn()", but based on the IBM SkillsBuild AI course.

I will divide the AI course into fragments. The game will contain multiple types of challenges (e.g., simulations, puzzles, building tasks) to teach different AI concepts. The specific design will be based on the IBM SkillsBuild course content.

An LLM tutor (IBM Granite) will support players when they are stuck, have questions, or want to learn extended knowledge based on the current challenge.

5. What is the PURPOSE of this project or why/to whom is this software/solution valuable? Software should not be the sole objective of the project but a means to allow your users/clients to achieve something that was not previously possible. If there is already a solution for your problem, why not use it? It is important that you know what this objective is from the outset. *

Different learning methods work for different people. This solution provides a method for people who like to play games and have difficulty staying focused on traditional online courses. Existing educational games do not have LLM support, and existing LLM tutors do not have engaging game mechanics. This project combines both. For IBM, this game can also help more people engage

6. What might you be covering in the LITERATURE REVIEW chapter of your project? BSc projects require a literature/background review. The importance and depth of such chapters vary from project to project. *

It will include these types of paper:

Game-Based Learning, AI or Computing Education, LLMs in Education, Educational Game Design, Evaluation Methods

7. How do you plan to EVALUATE your work? Note that I am not asking how you will test your software as that is something entirely different. Your report will need an evaluation chapter in which you assess to what extent you achieved your objective. The importance and depth of this chapter vary too from project to project. * 

User testing with 15-20 participants

- Pre/post AI knowledge quiz (8-10 questions based on SkillsBuild content)
- User experience questionnaire
- Open-ended feedback for improvement

Statistical methods (means, standard deviations) and thematic analysis of

8. What do you hope to ACHIEVE or ACCOMPLISH by doing this project? For example: perhaps you feel this is the way you achieve a particular objective, you might find supervised-proposed projects too hard or too research-focused and are looking for a more familiar challenge, you might wish to start a business based on an app and use this as an opportunity to build a prototype. Anything goes. All I ask is that you are completely honest with me (and yourself). * 

I hope to make an engaging educational game with Unity and publish it on the Steam platform. Through this project, I will learn front-end and back-end development, Unity, and LLM integration, and these skills are valuable for my future. I also hope to combine my psychology background with computing to create something that is both technically interesting and educationally meaningful.

9. Other comments/doubts/questions? 

輸入您的答案

10. Project's Zotero library link *

https://www.zotero.org/groups/6390740/individual_project_ibm_chaoyang/library

11. Preliminary report Overleaf link *

<https://www.overleaf.com/read/jdrzqgwszrcp#d1f5c9>



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